

Comprehensive Medical Record

SYNTHETIC SAMPLE — NOT A REAL PATIENT RECORD

Patient Information

Field	Details
Patient Name	Rahul Mehta
Patient ID	SYN-MED-2026-00421
Age / Sex	34 years / Male
Date of Birth	1992-04-14
Blood Group	O Positive
Height	171 cm
Weight	74 kg
Report Period	January 2026 – September 2026

Clinical Summary

This synthetic longitudinal record describes a fictional 34-year-old male who presented with intermittent fatigue, occasional dry cough, mild seasonal nasal congestion, and a family history of type 2 diabetes. The patient has no documented smoking or alcohol history in this sample. The record contains outpatient visits, laboratory investigations, imaging, pulmonary testing, medications, and follow-up observations over several months.

Visit 1 — Initial Consultation

Date: 2026-01-18

The patient reported fatigue for approximately three weeks and occasional difficulty concentrating during the afternoon. No fever, chest pain, fainting, or acute shortness of breath was reported. Physical examination in this fictional record was documented as generally unremarkable.

Initial plan: CBC, fasting glucose, HbA1c, thyroid profile, liver function, kidney function, and lipid profile.

Laboratory Panel — 2026-01-18

Investigation	Result	Unit	Reference Range
Hemoglobin	14.1	g/dL	13.0–17.0
WBC	7.8	thousand/ μ L	4.0–11.0
Platelets	262	thousand/ μ L	150–450
ESR	16	mm/hr	0–15
Fasting Glucose	104	mg/dL	70–99
HbA1c	5.8	%	<5.7
TSH	3.2	μ IU/mL	0.4–4.5
ALT	31	U/L	7–56
AST	26	U/L	10–40
Creatinine	0.9	mg/dL	0.6–1.2
eGFR	105	mL/min/1.73m ²	>90

Investigation	Result	Unit	Reference Range
Total Cholesterol	192	mg/dL	<200
LDL	118	mg/dL	<100 optimal
HDL	46	mg/dL	>40
Triglycerides	141	mg/dL	<150

Record note: The synthetic record flags fasting glucose, HbA1c, LDL, and ESR relative to the listed reference/target values. These values are included for RAG retrieval testing and do not establish a diagnosis.

Visit 2 — Respiratory Symptoms

Date: 2026-03-06

The patient reported intermittent dry cough and nasal congestion that became more noticeable during seasonal weather changes. No smoking history was documented. Examination was recorded as showing mild nasal congestion without acute respiratory distress.

Pulmonary Function Test — 2026-03-06

Parameter	Result	Reference / Note
FVC	4.12 L	Expected range for sample
FEV1	3.36 L	Expected range for sample
FEV1/FVC	81.6%	Preserved ratio
MMEF 25–75%	69%	Reduced relative to laboratory reference
Oxygen Saturation	98%	Room air

Radiology — Chest X-Ray: Cardiomedial silhouette within normal limits. No focal air-space opacity, pleural effusion, or pneumothorax identified in this synthetic report. Impression: no acute cardiopulmonary abnormality.

Clinical note: The fictional clinician documented possible allergic/airway irritation and recommended trigger avoidance and follow-up if symptoms persisted.

Medication Record

Date	Medication	Dose	Duration	Reason / Note
2026-03-06	Formoterol/Budesonide inhaler	As prescribed	15 days	Synthetic example for airway symptoms
2026-03-06	Saline nasal spray	As directed	As needed	Nasal congestion

Visit 3 — Metabolic Follow-Up

Date: 2026-05-14

The patient reported increased sedentary work hours and reduced regular exercise. No acute symptoms were reported. The fictional clinician advised regular physical activity, balanced nutrition, and repeat metabolic testing.

Investigation	January	May	Unit
Fasting Glucose	104	101	mg/dL
HbA1c	5.8	5.7	%
Total Cholesterol	192	185	mg/dL
LDL	118	109	mg/dL
HDL	46	48	mg/dL
Triglycerides	141	126	mg/dL

The May values show the direction of change across the synthetic record. They are provided to let a RAG system answer questions such as 'How did HbA1c change between January and May?' without inventing values.

Visit 4 — General Follow-Up

Date: 2026-07-22

The patient reported improved energy after establishing a more regular sleep schedule and walking routine. Cough was described as occasional. No new significant symptoms were documented in this fictional visit.

Repeat Laboratory Results — 2026-07-22

Investigation	Result	Unit	Reference Range
Hemoglobin	14.0	g/dL	13.0–17.0
WBC	7.4	thousand/ μ L	4.0–11.0
ESR	11	mm/hr	0–15
Fasting Glucose	97	mg/dL	70–99
HbA1c	5.5	%	<5.7
ALT	27	U/L	7–56
AST	23	U/L	10–40
Creatinine	0.9	mg/dL	0.6–1.2
Total Cholesterol	178	mg/dL	<200
LDL	102	mg/dL	<100 optimal
HDL	50	mg/dL	>40
Triglycerides	118	mg/dL	<150

Visit 5 — September Review

Date: 2026-09-15

The patient attended a routine follow-up. The record states that respiratory symptoms were infrequent. The patient continued a walking routine and was advised to maintain follow-up and repeat investigations when clinically appropriate.

Current Medication and Allergy Record

Category	Record
Current medications	No continuous medication listed in this synthetic September record.
Previous short course	Synthetic inhaler course documented in March 2026.
Known drug allergies	None documented in this synthetic record.
Smoking	Not documented as a current habit in this synthetic record.
Alcohol	Not documented as a current habit in this synthetic record.

Longitudinal Summary

Across the fictional 2026 record, the patient's CBC and kidney/liver markers remained within the stated ranges. Metabolic markers changed over time: HbA1c moved from 5.8% in January to 5.7% in May and 5.5% in July; LDL moved from 118 mg/dL to 109 mg/dL and then 102 mg/dL. The respiratory work-up included a chest X-ray with no acute finding and a PFT showing an MMEF value of 69% in the synthetic report.

Useful RAG Questions for This Document

- What is the patient's current HbA1c?
- How did the HbA1c change from January to July 2026?

- What were the patient's LDL values across the year?
- What did the chest X-ray report say?
- What was the MMEF result on the pulmonary function test?
- What medications were recorded in March 2026?
- What was the patient's creatinine on the January and July tests?
- Summarize the patient's medical history for 2026 using only this report.
- Which investigations were repeated during follow-up?
- What symptoms were documented during the March respiratory visit?

SYNTHETIC DATA NOTICE: This entire document was generated for software/RAG development and evaluation. It contains fictional patient information and must not be used for clinical decision-making.